

A Hybrid Edge-Cloud IoT Wearable for HRV-Based Fatigue and Safety Monitoring for Outdoor Infrastructure Workers

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ABSTRACT

Construction workers face safety risks such as fatigue, abnormal vital signs, and fall-related incidents. This report presents a hybrid edge–cloud IoT wearable prototype for real-time health and safety monitoring in high-risk construction environments. The prototype integrates an ESP32-C3 microcontroller, a MAX30102 heart-rate and SpO₂ sensor, an MPU6050 motion sensor, a local buzzer, a cancel button, and Wi-Fi/MQTT communication with AWS IoT Core.

At the edge level, the device performs sensor acquisition, physiological screening, motion-based fall-risk detection, and HRV-based fatigue analysis using BPM-derived inter-beat interval approximation. SDNN and RMSSD are extracted over analysis windows to support fatigue-warning logic, while abnormal SpO₂, heart-rate, and motion conditions trigger local buzzer alerts. At the cloud level, telemetry and alert payloads are sent to AWS IoT Core for rule-based processing, storage, dashboard visualization, and notification support.

By combining immediate local warning with cloud-assisted supervision, the system demonstrates a functional proof-of-concept for continuous worker safety monitoring and preventive risk management on construction sites.

Keywords: Construction safety; Wearable IoT; Heart rate variability; Fatigue monitoring; Fall-risk detection; PM2.5 monitoring; Edge–cloud architecture; AWS IoT.

1 Introduction

1.1 Problem and SDG Context

Problem Statement. Vietnam’s construction sector is expanding rapidly, with the market projected to reach \$110.42 billion by 2030 [1]. However, outdoor construction work remains physically demanding and hazardous. Workers are exposed to fall risks, heat stress, fine dust, abnormal vital-sign conditions, and fatigue-related performance decline [2]. In many construction sites, worker safety still depends mainly on manual supervision and delayed incident reporting, which makes it difficult to detect early physiological deterioration or respond quickly to sudden events. In Vietnam, workplace accidents have also caused substantial economic losses, with reported losses exceeding VND 43 trillion in 2024 [3].

A wearable IoT approach is therefore necessary because it enables continuous health and motion monitoring directly at the worker level. By combining edge-based risk detection with cloud-based supervision, the proposed system can support earlier warning, faster local response, and better safety visibility for site managers.

Target SDGs. The Smart Safety IoT Wristband contributes mainly to two United Nations Sustainable Development Goals:

- **SDG 3 (Good Health and Well-being):** The system supports preventive worker health protection by monitoring heart rate, SpO₂, HRV-based fatigue indicators, and fall-risk events. Local buzzer alerts and cloud telemetry help reduce response delay when abnormal physiological or motion conditions are detected.
- **SDG 11 (Sustainable Cities and Communities):** The system contributes to safer urban infrastructure development by giving supervisors real-time visibility of worker safety conditions on construction sites. This supports safer, more resilient, and more humane construction practices.

Table 1: Technical contribution of the proposed IoT system to the selected SDGs

Target SDG	Humanitarian Issue	IoT Technical Contribution
SDG 3	Preventable injury, fatigue-related risk, hypoxemia, fall events, and delayed emergency response among outdoor construction workers	Implemented wearable monitoring of heart rate, SpO ₂ , HRV fatigue indicators, and motion-based fall-risk events; local buzzer alerting and MQTT-based cloud telemetry for supervisor monitoring
SDG 11	Safety risks in urban infrastructure construction and maintenance environments	Real-time worker-status data for dashboard-based supervision, risk tracking, and safer infrastructure worksite management. PM _{2.5} sensing and SIM768x-based 4G emergency notification are planned or simulated extensions in the current prototype.

1.2 Main Contributions

This work presents a functional prototype of a hybrid edge–cloud wearable system for construction worker safety monitoring. The main contributions are summarized as follows:

- A wearable edge device is implemented using an ESP32-C3 microcontroller, a MAX30102 physiological sensor, an MPU6050 inertial sensor, a local buzzer, and a cancel button for on-device safety monitoring.
- An edge-level risk screening mechanism is developed to detect abnormal heart rate, low blood oxygen saturation, fatigue-related HRV patterns, and motion-based fall-risk events before cloud transmission.
- A cloud-assisted monitoring pipeline is integrated using MQTT over TLS, AWS IoT Core, Lambda, DynamoDB, S3, and SNS to support remote supervision, data logging, and alert notification.
- A controlled demonstration is conducted with representative normal, danger, fall-emergency, and exhaustion-risk scenarios to verify the end-to-end edge–cloud workflow.

2 Related Work

Wearable sensing has been widely explored for occupational safety because it allows physiological and motion data to be collected close to the worker. Consumer smartwatches can provide heart-rate monitoring, activity tracking, and in some cases fall detection. However, they are not specifically designed for construction-site operation, where dust, vibration, heat exposure, and safety-supervision requirements are critical.

Vision-based systems such as CCTV and computer-vision monitoring can support site-level hazard detection. Nevertheless, camera-based approaches cannot directly observe internal physiological conditions such as hypoxemia, abnormal heart rate, or fatigue-related autonomic changes. They may also be limited by occlusion, lighting conditions, and privacy concerns.

Accelerometer-based fall detectors provide a lightweight solution for sudden motion events. Most basic systems, however, focus mainly on fall detection and do not combine physiological monitoring, local warning, and cloud-based supervisor notification in a single workflow.

HRV-based fatigue monitoring has also been studied as a non-invasive indicator of autonomic nervous system activity. In practical embedded systems, however, reliable HRV estimation requires careful signal acquisition and noise handling. In the current

Table 2. Comparison between the proposed prototype and common safety-monitoring approaches

Approach	HR/SpO ₂	HRV	Fall	Local Alert	Cloud	Construction Focus
Manual supervision	No	No	Limited	No	No	Partial
Consumer smartwatch	Yes	Partial	Partial	Yes	Yes	No
CCTV monitoring	No	No	Partial	No	Yes	Partial
Basic fall detector	No	No	Yes	Yes	Limited	No
Proposed prototype	Yes	Yes	Yes	Yes	Yes	Yes

prototype, HRV is approximated from BPM-derived inter-beat intervals to demonstrate the workflow. Production-grade fatigue analysis should use beat-to-beat PPG timestamps.

Compared with these approaches, the proposed system focuses on construction-specific safety monitoring by combining physiological sensing, motion-based fall-risk detection, HRV-based fatigue screening, local buzzer alerting, and cloud-assisted supervision.

3 Market Intelligence and Problem Scale

To quantify the potential market demand for the Smart Safety IoT Wristband, a top-down analytical approach is applied using the TAM/SAM/SOM model. The baseline workforce assumptions are derived from national labour statistics and construction-sector reports. Vietnam recorded approximately 51.9 million employed workers in 2024 [4]. In addition, the large informal employment segment, reported at about 33 million workers in Q3/2024, indicates that the serviceable market should focus on formal or managed construction workers rather than the entire labour force [5].

Core Data Sources: The analysis uses labour-market data from the General Statistics Office of Vietnam, informal-employment statistics reported by VietnamPlus/VNA, Vietnam’s occupational safety and hygiene law, and a construction-sector market report.

3.1 TAM (Total Addressable Market) Analysis

- **Market Analysis:** According to national labour statistics, Vietnam recorded approximately 51.9 million employed workers in 2024 [4]. For this analysis, the construction-only workforce is estimated by applying a 25–30% allocation to the broader industry-and-construction labour segment. A midpoint estimate of **4.8 million construction workers** is therefore used as the TAM worker base. **4.8 million construction workers** is therefore used as the TAM worker base.

- **Quantitative Formula:**

$$\text{TAM Value} = N (\text{total construction workers}) \times E (\text{average basic safety-equipment spending})$$

- **Estimation:** Assuming an average basic safety-equipment spending of 5,000,000 VND per worker annually, the estimated TAM value is:

$$4,800,000 \times 5,000,000 = 24,000 \text{ billion VND.}$$

This is equivalent to approximately USD 1 billion, depending on the exchange-rate assumption.

3.2 SAM (Serviceable Available Market) Analysis

- **Market Analysis:** The SAM narrows the market to formal or managed construction workers who are more likely to operate under structured occupational safety requirements. This filtering is justified by Vietnam’s large informal labour segment, which was reported at approximately 33 million workers, accounting for 63.9% of total employment in Q3/2024 [5]. In addition, Vietnam’s occupational safety and hygiene law requires risk assessment and emergency-rescue planning for workplaces with occupational hazards [6].

- **Quantitative Formula:**

$$\text{SAM Value} = N (\text{formal or managed workers}) \times P (\text{premium safety and IoT equipment spending})$$

- **Estimation:** Assuming a serviceable formal/managed construction segment of 780,000 workers and a premium safety-and-IoT

spending level of 20,000,000 VND per worker, the estimated SAM value is:

$$780,000 \times 20,000,000 = 15,600 \text{ billion VND.}$$

3.3 SOM (Serviceable Obtainable Market) Analysis

- **Market Analysis:** In the initial commercialization phase, the project targets large general contractors and managed construction sites, where safety compliance, centralized procurement, and cloud-based monitoring are more feasible. The construction sector remains economically significant, contributing approximately 6–7% of Vietnam’s GDP during 2013–2023 [7]. Based on this target segment, a practical SOM of **250,000 workers** is assumed.

- **Quantitative Formula:**

$$\text{SOM Value} = N \text{ (target contractor workers)} \times C \text{ (IoT wristband service package cost)}$$

- **Estimation:** Assuming a first-year IoT safety service package price of 8,000,000 VND per worker, including the wearable device, cloud dashboard access, data storage, maintenance, and technical support, the practical SOM value is:

$$250,000 \times 8,000,000 = 2,000 \text{ billion VND.}$$

4 Strategic Analysis

To evaluate the competitive position of the Smart Safety IoT Wristband, this section compares the proposed system with existing safety-monitoring alternatives and summarizes its strategic value for construction safety.

Alternative Comparison and Unique Value. Compared with manual supervision, the system provides continuous worker-level monitoring and immediate local alerts. Compared with consumer smartwatches, it is designed for construction safety by integrating HR/SpO₂ monitoring, HRV-based fatigue indicators, fall-risk detection, and cloud telemetry. Compared with CCTV, it can detect physiological risks such as low SpO₂, abnormal heart rate, and fatigue, which are not visible through cameras. Compared with basic fall detectors, it supports both early warning and emergency response.

SWOT Summary. The system’s main strengths are edge-based risk detection, local buzzer alerts, and cloud-based supervisor visibility. Current limitations include prototype size, power consumption, and the fact that PM_{2.5} sensing and SIM768x-based 4G emergency communication remain planned or simulated extensions. Key opportunities come from Vietnam’s growing construction sector, Industry 4.0 adoption, and stricter safety requirements in FDI and public infrastructure projects. Main adoption risks include worker resistance, privacy concerns, harsh outdoor conditions, dust, heat, vibration, and hardware durability.

Overall, the Smart Safety IoT Wristband is positioned as a construction-specific safety-monitoring solution rather than a general consumer wearable. Its key value lies in combining early warning, emergency response, real-time supervisor visibility, and traceable safety records in a scalable platform.

STRENGTHS • Advanced Technical Capabilities ◦ Features cost-effective Wi-Fi 6, edge computing for noise filtration, and high-fidelity sensors.	WEAKNESSES • Prototype Constraints ◦ Current hardware maintains a bulky form factor and relies heavily on site-wide Wi-Fi.
OPPORTUNITIES • Market Growth Opportunities ◦ Fueled by stricter safety regulations, BIM integration, and rapid mega-project expansion.	THREATS • External Adoption Threats ◦ Facing user resistance, low-cost smartwatch competition, and extreme physical site environments.

Figure 1: SWOT analysis matrix for the proposed system

5 Business Model and Sustainability

5.1 Market Validation

Preliminary market analysis suggests that Tier-1 contractors may have strong demand for continuous safety-monitoring solutions, because the financial consequences of fatal accidents, project suspension, and legal liability can exceed the cost of preventive safety equipment.

5.2 Strategic Framework: Business Model Canvas

As shown in Figure 2, the business model follows a B2B approach targeting Tier-1 contractors that hold legal responsibility for site safety. The main value proposition is continuous worker monitoring, local warning, cloud-based supervisor visibility, and reduced accident-liability risk. Revenue is generated through initial device sales or leasing and recurring SaaS-based dashboard, maintenance, and support fees.

KEY PARTNERSHIPS Suppliers, AWS, and Safety Agencies Strategic alliances with hardware component suppliers and cloud infrastructure providers.	KEY ACTIVITIES R&D and AWS Ops Focused on hardware innovation and robust cloud operations.	KEY RESOURCES Sensors and Hardware Integration of MAX30102 sensors and hardware components.	VALUE PROPOSITIONS 24/7 Monitoring and Liability Reduction Providing real-time SOS alerts and safety data to protect workers and minimize corporate risk.	CUSTOMER SEGMENTS Top 10 Contractors Targeting large-scale construction contractors.
	CUSTOMER RELATIONSHIPS Tech Support Maintaining long-term support relationships.		CHANNELS Direct Sales & Trade Shows Reaching target contractors via direct sales and trade shows.	
COST STRUCTURE 1.2M BOM Costs associated with R&D, AWS, and Bill of Materials.			REVENUE STREAMS 2.5M VND Price Revenue driven by device sales, leasing, and recurring SaaS fees.	

Figure 2: Business Model Canvas for Smart Safety IoT System

5.3 Price Structure and Bill of Materials

The device retail price is estimated at 2,000,000 VND per unit. For commercial deployment, the first-year service package is estimated at 8,000,000 VND per worker, including the device, cloud dashboard access, data storage, maintenance, and technical support. Therefore, the 2,000,000 VND figure represents the hardware acquisition price, while the 8,000,000 VND figure represents the full first-year safety-monitoring service package. And to eliminate false alarms in harsh environments, high-fidelity components are used. The prototype BOM is strictly optimized at **1,019,000 VND**:

- MAX30102 heart-rate and SpO₂ sensor: 648,000 VND;
- XIAO ESP32-C3 microcontroller: 216,000 VND;
- Li-Po battery and charging circuit: approximately 80,000 VND;
- MPU6050 accelerometer and gyroscope: 75,000 VND.

5.4 Financial Sustainability Justification

With a BOM of 1,019,000 VND, the retail price is set at **2,000,000 VND/device**, yielding a ~48% gross margin to sustain AWS cloud operations and R&D. While the premium MAX30102 sensor increases costs, it prevents disruptive false alarms during heavy labor. For contractors, a 2.5 million VND investment per worker delivers immense ROI by mitigating the risk of massive legal liabilities. Mass production will further reduce component costs by 20-30%, ensuring long-term economic resilience.

6 System Overview

The proposed hybrid edge-cloud IoT wearable monitors health, fatigue, and safety of outdoor construction workers. It collects physiological and motion data, assesses risks at the edge, provides local warnings, and transmits telemetry to the cloud for supervisor oversight, reducing response delay in hazardous sites.

The current prototype uses a XIAO ESP32-C3, MAX30102 (heart rate, SpO₂), MPU6050 (acceleration, gyroscope), an active

Table 3. Implementation status of the prototype functions

Function	Status	Evidence
Heart-rate and SpO ₂ sensing	Implemented	MAX30102 sensor
Motion and fall-risk sensing	Implemented	MPU6050 sensor
Local warning	Implemented	GPIO-controlled buzzer
Alarm cancellation	Implemented	Pull-up cancel button
MQTT telemetry	Implemented	AWS IoT Core test client
Cloud classification	Implemented	Lambda processing
Data storage	Implemented	DynamoDB and S3
Supervisor notification	Implemented	SNS email for danger/emergency
PM _{2.5} sensing	Planned / partial extension	APM10 node or placeholder value
SIM768x emergency call/SMS	Simulated	Serial/cloud alert logic
HRV fatigue analysis	Demonstrated	BPM-derived IBI approximation

buzzer, a cancel button, and Wi-Fi/MQTT. When a warning or emergency is detected, the edge activates the buzzer and publishes telemetry or alerts to AWS IoT.

The system has two operating paths: an edge-only path for immediate local alerting, and a cloud path for remote monitoring via Wi-Fi/MQTT. This hybrid design handles urgent events locally while maintaining cloud visibility for site management.

PM_{2.5} sensing, cloud-to-device threshold updates, and SIM768x-based 4G/SMS emergency communication are planned or simulated extensions, not physically implemented in the current prototype.

The prototype is considered successful if it can demonstrate the following core functional objectives:

- acquire real-time heart-rate and SpO₂ data from the MAX30102 sensor;
- process motion data from the MPU6050 for abnormal movement and fall-risk detection;
- estimate fatigue-related risk using HRV indicators derived from heart-rate data;
- activate the local buzzer when a warning or emergency condition is detected;
- transmit telemetry and alert data to AWS IoT services through Wi-Fi/MQTT;
- demonstrate the intended emergency-notification workflow through simulated alert logic.

6.1 Prototype Implementation Scope

7 System Architecture

The proposed system adopts a hybrid edge–cloud architecture for wearable worker safety monitoring. As shown in Figure 3, the architecture spans from physical sensing at the worker level to cloud-based processing and supervisor monitoring. This layered structure allows urgent safety events to be handled locally while still preserving cloud visibility for logging, analytics, and notification.

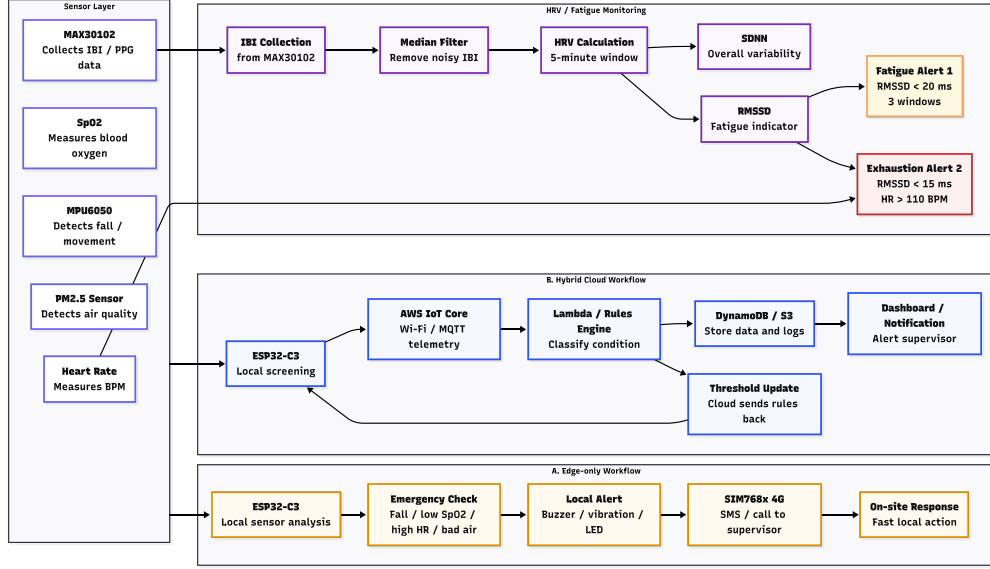


Figure 3: Hybrid edge–cloud system architecture and workflow of the worker safety monitoring prototype. The current implementation uses ESP32-C3, MAX30102, MPU6050, buzzer alerting, and Wi-Fi/MQTT telemetry. PM_{2.5} sensing, SIM768x-based 4G notification, and cloud-to-device threshold updates are planned or simulated extensions.

7.1 Architecture Layers

The architecture consists of five layers:

- **Perception layer:** MAX30102 (heart rate, SpO₂) and MPU6050 (acceleration, gyroscope) collect worker data. PM_{2.5} is a planned extension.
- **Edge-processing layer:** The ESP32-C3 performs local screening, feature extraction (HRV, SVM), and threshold-based risk evaluation before cloud transmission.
- **Local response layer:** The buzzer activates immediately on abnormal conditions, providing fast on-site warning even without cloud connectivity.
- **Communication and cloud-ingestion layer:** The device publishes telemetry, HRV summaries, and alerts to AWS IoT Core via Wi-Fi and MQTT over TLS, using separate topics.
- **Cloud-processing and application layer:** AWS IoT Core forwards messages to Lambda for classification, DynamoDB storage, S3 logging, and dashboard or notification outputs.

7.2 Edge-Only and Hybrid Operating Paths

The system supports two complementary paths. The **edge-only path** handles urgent local response: sensor data are analysed on the ESP32-C3, abnormal conditions are screened locally, and the buzzer activates without depending on cloud availability.

The **hybrid cloud path** provides remote monitoring and traceability. After local processing, the device sends MQTT payloads to AWS IoT Core. The cloud classifies the worker condition, stores records in DynamoDB and S3, and sends notifications for danger or emergency cases, supporting supervisor visibility and analytics.

7.3 HRV and Fatigue Monitoring Path

The HRV fatigue path uses IBI approximated from heart rate (IBI = 60000/BPM). IBI samples are filtered and accumulated over a five-minute window, then SDNN and RMSSD are computed. RMSSD is the main fatigue indicator: a level-1 warning is generated when RMSSD remains below 20 ms for three consecutive windows; exhaustion risk (level 2) when RMSSD falls below 15 ms and heart rate exceeds 110 BPM.

In the current prototype, HRV processing demonstrates the intended workflow, but production should use beat-to-beat PPG timestamps rather than BPM-based IBI approximation.

7.4 Communication Protocols and Interactions

The main protocol is MQTT over TLS through AWS IoT Core. The device publishes three message types: telemetry, HRV summaries, and alerts. The cloud receives these, applies rule-based processing, stores records, and sends notifications when required. This design combines immediate local response with cloud-assisted monitoring, giving supervisors access to real-time and historical worker data.

7.5 ESP-NOW Local Communication for Environmental Sensing

To incorporate air-quality data without exposing the environmental node to the internet, the system uses ESP-NOW, a lightweight, connectionless protocol from Espressif. An ESP32 node equipped with an APM10 UART sensor reads PM1.0, PM2.5, and PM10 values every 2 seconds, computes a prototype AQI from PM2.5, and transmits a packed `EnvPacket` directly to the ESP32-C3 gateway. The packet includes a magic number, version, sequence number, PM values, AQI, and uptime. Both devices must share the same Wi-Fi channel (e.g., channel 13), and the node must be pre-configured with the gateway’s MAC address. This design keeps the node lightweight and power-efficient; the gateway merges the environmental data with biometric and motion telemetry before forwarding the combined payload to AWS IoT Core via MQTT.

8 Hardware and Software Design

This section describes the hardware and embedded firmware of the implemented prototype. The design follows an edge-first safety approach: sensor acquisition, signal screening, risk classification, and local alarm activation are performed on the device before data are transmitted to the cloud, ensuring safety-critical warnings remain available even when cloud connectivity is delayed or unavailable.

8.1 Hardware Components

The prototype is built around a XIAO ESP32-C3, coordinating sensing, processing, alarm control, and Wi-Fi/MQTT. Physiological monitoring uses a MAX30102; motion monitoring uses an MPU6050 [8, 9]. An active buzzer provides local warnings, and a cancel button silences the alarm without interrupting monitoring. Table 4 summarises the main components.

Table 4: Main hardware components of the implemented wearable prototype

Component	Model / Interface	Function
Microcontroller	XIAO ESP32-C3	Main Wi-Fi-enabled edge controller for sensing, processing, alarm control, and MQTT communication.
Physiological sensor	MAX30102, I2C	Measures heart rate and SpO ₂ for biometric monitoring and HRV-based fatigue estimation.
Motion sensor	MPU6050, I2C	Measures acceleration and gyroscope data for abnormal motion and fall-risk detection.
Local actuator	Active buzzer	Provides immediate audible warning when a risk condition is detected.
User input	Cancel button	Allows manual alarm cancellation during testing without interrupting monitoring or telemetry.
Wireless communication	ESP32-C3 Wi-Fi	Supports MQTT-based telemetry and alert transmission to AWS IoT services.
Environmental node	ESP32S NodeMCU + APM10 (UART)	Reads PM1.0, PM2.5, PM10 via UART2 (GPIO16/17, 1200 baud), computes AQI, sends to gateway via ESP-NOW.
Emergency communication	SIM768x 4G module, simulated	Intended for future SMS or voice-call notification; currently represented by simulated alert logic.

8.2 Environmental Sensing Node

To extend the system with air-quality monitoring, a dedicated ESP32 node (ESP32S NodeMCU) is connected to an APM10 UART particulate-matter sensor. The sensor communicates via UART2 (GPIO16 as RX, GPIO17 as TX) at 1200 baud. The firmware periodically sends a read command `0xFE`, `0xA5`, `0x00`, `0x01`, `0xA6` and decodes the returning frame to extract PM1.0, PM2.5, and PM10 values (big-endian 16-bit fields). A prototype AQI is calculated from PM2.5 using US EPA breakpoint interpolation, with PM2.5 truncated to one decimal place. Checksum verification is disabled by default to accommodate sensor firmware variants, making the AQI suitable for demonstration rather than regulatory use. The node then packages the data into an ESP-NOW packet and transmits it to the gateway every two seconds.

The environmental node uses the following UART pins on its ESP32S:

- APM10 TX → ESP32S GPIO16 (UART2 RX)
- APM10 RX → ESP32S GPIO17 (UART2 TX)

Figure 4 shows the proposed hardware schematic. In the current submitted prototype, the ESP32-C3, MAX30102, MPU6050, buzzer, cancel button, and Wi-Fi/MQTT communication are physically implemented. PM_{2.5} sensing and SIM768x-based 4G communication are included in the proposed design but remain planned or simulated extensions.

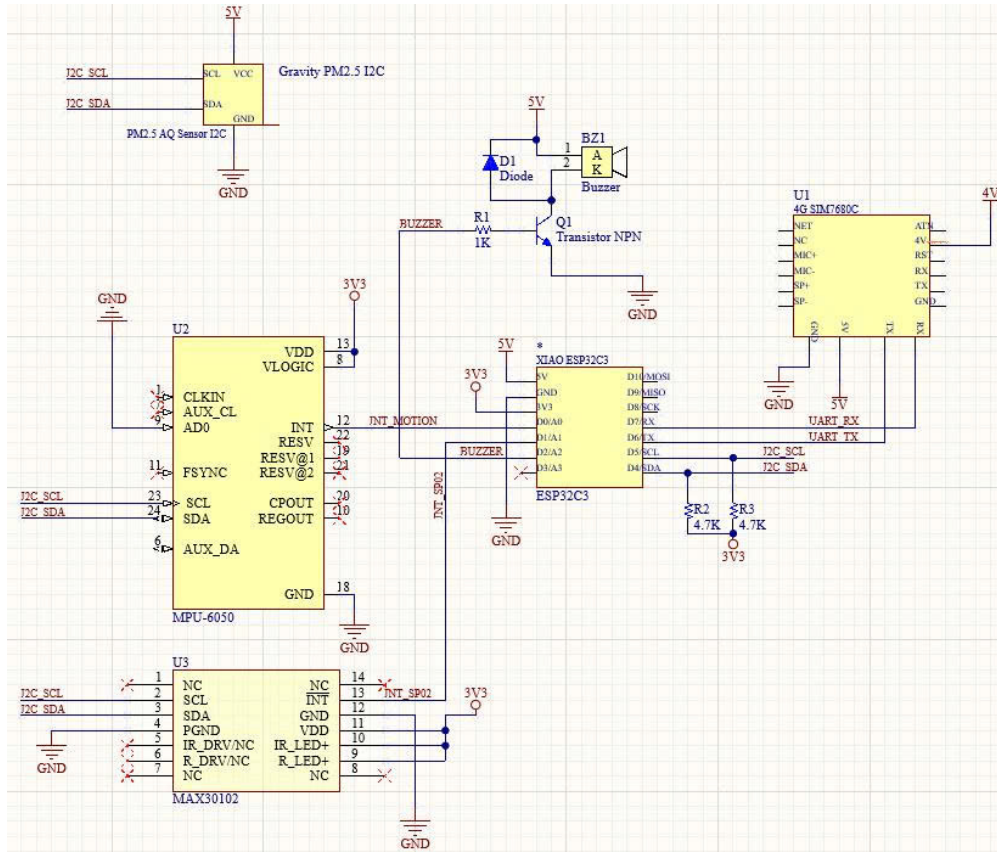


Figure 4: Hardware schematic of the proposed wearable system. The current prototype physically implements the ESP32-C3, MAX30102, MPU6050, buzzer, cancel button, and Wi-Fi/MQTT communication, while PM_{2.5} sensing and SIM768x-based 4G communication are planned or simulated extensions.

The MAX30102 and MPU6050 modules share the I2C bus, with SDA and SCL assigned to GPIO6 and GPIO7, respectively. The buzzer is connected to GPIO2, while the cancel button is connected to GPIO3 using an internal pull-up configuration. Table 5 summarizes the pin assignment used in the firmware.

Table 5: Pin assignment of the implemented wearable prototype

Table 6. Main software libraries used in the embedded firmware

Library	Purpose
Wire.h	Provides I2C communication for the MAX30102 and MPU6050 sensors.
DFRobot_BloodOxygen_S.h	Supports heart-rate and SpO ₂ acquisition from the MAX30102 sensor.
Adafruit_MPU6050.h Adafruit_Sensor.h	Support acceleration and gyroscope data acquisition from the MPU6050 module.
WiFi.h WiFiClientSecure.h	Provide Wi-Fi connectivity and TLS-secured communication for AWS IoT.
PubSubClient.h	Enables MQTT communication with AWS IoT Core.
ArduinoJson.h	Constructs structured JSON payloads for telemetry, HRV summaries, and alert events.
FreeRTOS headers	Support task scheduling, queues, mutexes, and non-blocking firmware execution.

Module / Signal	ESP32-C3 Pin	Purpose
MAX30102 SDA	GPIO6	I2C data line for physiological sensing
MAX30102 SCL	GPIO7	I2C clock line for physiological sensing
MPU6050 SDA	GPIO6	Shared I2C data line for motion sensing
MPU6050 SCL	GPIO7	Shared I2C clock line for motion sensing
Active buzzer	GPIO2	Local audible alarm output
Cancel button	GPIO3	Manual alarm cancellation using internal pull-up input
MAX30102 address	0x57	I2C address of the physiological sensor

8.3 Embedded Firmware Architecture

The firmware is written in Arduino C++ with FreeRTOS tasks, queues, and mutexes. It separates sensing, alarm handling, and cloud communication into concurrent tasks, improving responsiveness and preventing network delays from blocking local safety processing.

Three main tasks run:

- **Anomaly Detection (highest priority):** reads sensors, validates data, computes motion/HRV features, evaluates risk thresholds, generates alerts.
- **Alarm Control:** controls buzzer and cancel button with non-blocking logic.
- **Cloud Publish:** maintains Wi-Fi/MQTT and publishes telemetry, HRV summaries, and alerts.

Alert events and HRV summaries use queues; shared sensor data are mutex-protected. This decouples real-time risk detection from network transmission, improving stability.

8.4 Software Libraries

The firmware uses libraries for I2C, sensor acquisition, secure networking, MQTT, JSON, and FreeRTOS. Wi-Fi and MQTT transmit telemetry and alerts to AWS IoT Core over TLS. Table 6 summarises the main libraries.

8.5 Data Acquisition and Feature Extraction

Sensor acquisition is handled by the anomaly detection task. Heart-rate and SpO₂ readings from the MAX30102 are screened for physiological plausibility before being used for risk classification [8]. Invalid SpO₂ values outside the expected range are rejected, and heart-rate values are accepted only within a predefined plausible range.

In the current implementation, valid heart-rate readings are used to estimate the inter-beat interval as:

$$IBI = \frac{60000}{BPM}$$

where IBI is expressed in milliseconds and BPM denotes the measured heart rate. This approach demonstrates the HRV workflow, but it remains an approximation. Production-grade HRV analysis should use beat-to-beat timestamps extracted directly from the PPG signal.

For motion analysis, acceleration data from the MPU6050 are used to compute the signal vector magnitude, a common acceleration-based feature for wearable fall-risk detection [10]:

$$SVM = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

where a_x , a_y , and a_z represent acceleration along the three axes. The SVM feature provides an orientation-independent representation of body motion and is used for free-fall, impact, and fall-risk detection.

8.6 HRV-Based Fatigue Analysis

Fatigue monitoring is performed using a five-minute HRV analysis window. Valid IBI samples are filtered using a median filter to reduce transient noise before being stored in the HRV buffer. Two HRV indicators are computed:

$$SDNN = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (IBI_i - \overline{IBI})^2}$$

$$RMSSD = \sqrt{\frac{1}{N-1} \sum_{i=2}^N (IBI_i - IBI_{i-1})^2}$$

SDNN reflects overall heart-rate variability, while RMSSD is used as the main short-term HRV indicator [11]. The firmware generates a level-1 fatigue warning when RMSSD remains below 20 ms for three consecutive HRV windows. A level-2 exhaustion-risk alert is generated when RMSSD falls below 15 ms while heart rate exceeds 110 BPM.

8.7 Risk Classification Logic

The edge-level risk logic combines physiological, fatigue-related, and motion-based indicators. Physiological risk is evaluated using heart-rate and SpO_2 thresholds. Fatigue risk is evaluated using RMSSD-based rules. Fall-risk detection is implemented as a finite-state sequence consisting of free-fall detection, impact detection, and post-impact confirmation. Table 7 summarizes the main thresholds used in the firmware.

When a warning or emergency condition is detected, the firmware activates the local buzzer and generates an alert event. The cancel button can silence the buzzer without stopping sensor acquisition, risk evaluation, or telemetry publishing.

8.8 Implementation Notes

The current firmware prioritizes stable demonstration of the core safety functions: MAX30102 physiological sensing, MPU6050 motion sensing, local buzzer alerting, cancel-button handling, FreeRTOS task execution, and MQTT-based telemetry publishing. $PM_{2.5}$ sensing is represented by a placeholder air-quality value because the dust-sensor library caused compatibility and stability issues during integration. The SMS function is also simulated through serial output and cloud alert payloads rather than physical cellular transmission. These limitations are addressed further in the Limitations and Future Work section.

9 Connectivity and Data Flow

This section explains the networking infrastructure and end-to-end data trajectory of the FOURUT wearable IoT system. The implemented prototype uses Wi-Fi and MQTT over TLS to transmit worker telemetry, HRV summaries, and alert events from

Table 7. Risk thresholds implemented in the edge firmware

Condition	Threshold	System Response
SpO ₂ warning	SpO ₂ < 90%	Generates a hypoxemia warning event.
Critical SpO ₂	SpO ₂ < 80%	Triggers a high-severity alarm and emergency alert.
Bradycardia warning	BPM < 50	Generates a physiological warning event.
Tachycardia warning	BPM > 120	Generates a physiological warning event.
Fatigue warning	RMSSD < 20 ms for three HRV windows	Generates a fatigue-warning event.
Exhaustion risk	RMSSD < 15 ms and BPM > 110	Triggers a fatigue-related emergency alert.
Free-fall phase	SVM < 4.9 m/s ²	Indicates possible loss of support.
Impact phase	SVM > 29.4 m/s ² within 1500 ms	Indicates possible impact following free-fall.
Post-impact confirmation	3000 ms waiting period after impact	Confirms a fall-risk emergency in the current prototype.

the ESP32-C3 edge device to AWS IoT Core. From AWS IoT Core, the data are forwarded to a Lambda function for cloud-side classification, storage, historical logging, and supervisor notification.

9.1 Communication Method

The ESP32-C3 edge device connects to the cloud through Wi-Fi and publishes structured JSON payloads using MQTT over TLS. MQTT is used because it is lightweight and suitable for IoT telemetry, while TLS-secured communication protects data in transit. In the submitted firmware, sensitive certificate fields are represented using placeholders so that private keys and credentials are not exposed in the public submission.

The edge device publishes three types of messages: periodic telemetry, HRV summaries, and alert events. These messages are separated by MQTT topic so that the cloud backend can distinguish routine monitoring data from fatigue-related summaries and emergency alerts.

9.2 MQTT Topic Structure

The MQTT topic structure follows the pattern:

`worker/{worker_id}/{message_type}`

For the current prototype worker ID W001, the implemented topics are summarized in Table 8.

Table 8: MQTT topics used by the wearable prototype

MQTT Topic	Purpose
<code>worker/W001/telemetry</code>	Publishes periodic worker status, including heart rate, SpO ₂ , SVM, HRV values, alarm state, and fall status.
<code>worker/W001/hrv</code>	Publishes HRV summaries such as SDNN, RMSSD, IBI count, heart rate, and alarm level.
<code>worker/W001/alert</code>	Publishes warning, danger, and emergency alert events detected by the edge device.

During testing, the AWS IoT MQTT test client can subscribe to `worker/#` to observe all worker-related messages under the same namespace.

9.3 End-to-End Data Flow

Figure 5 illustrates the complete data path from physical sensor acquisition to cloud-side storage and notification.

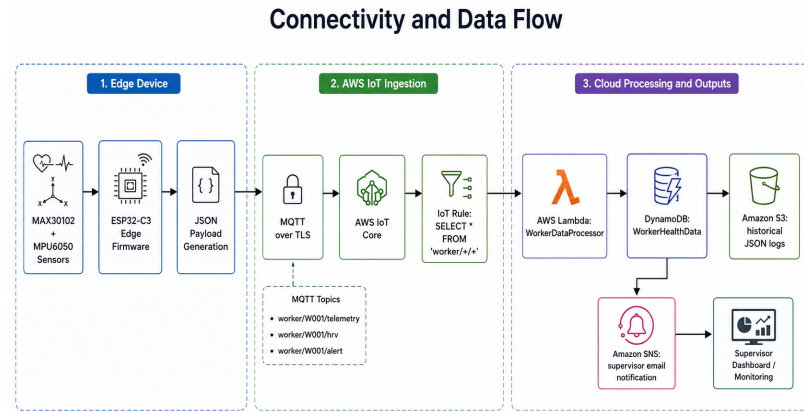


Figure 5: End-to-end connectivity and data flow from the ESP32-C3 edge device to AWS cloud services

The end-to-end data flow operates as follows:

- (i) The MAX30102 and MPU6050 sensors collect physiological and motion data.
- (ii) The ESP32-C3 firmware validates sensor readings, extracts safety-related features, and evaluates local risk conditions.
- (iii) The device generates JSON payloads for telemetry, HRV summaries, or alert events.
- (iv) The payloads are published to AWS IoT Core using MQTT over TLS.
- (v) An AWS IoT Rule captures worker-related messages using the SQL statement `SELECT * FROM 'worker/+/*'`.
- (vi) The IoT Rule forwards matching messages to the `WorkerDataProcessor` Lambda function.
- (vii) Lambda normalizes the event format, extracts the worker ID, classifies the danger level, and generates a processing result.
- (viii) Processed records are stored in DynamoDB, historical JSON logs are saved to S3, and SNS email notifications are sent for danger or emergency conditions.

9.4 Cloud Processing and Storage

The cloud backend uses a Lambda function named `WorkerDataProcessor` as the main processing unit. The function receives MQTT payloads forwarded by AWS IoT Core, normalizes the input format, extracts the worker ID, and classifies the worker condition into four levels: `NORMAL`, `WARNING`, `DANGER`, and `EMERGENCY`. The classifier uses both edge-generated indicators and sensor values, including heart rate, SpO_2 , AQI, RMSSD, alarm level, fall-detection status, alert category, and alert status.

The processed data are stored in the DynamoDB table `WorkerHealthData`. The table uses `DeviceID` as the partition key and `Timestamp` as the sort key, allowing multiple time-ordered records to be stored for each worker or device. In addition, the Lambda function can save historical JSON logs to Amazon S3 under the `worker-safety-logs/` prefix. For supervisor notification, Amazon SNS is used to send email alerts when the danger level is classified as `DANGER` or `EMERGENCY`.

Table 9: Cloud-side components and responsibilities

Cloud Component	Responsibility
AWS IoT Core	Receives MQTT messages from the ESP32-C3 edge device.
IoT Rule	Captures worker-related topics using <code>SELECT * FROM 'worker/+/+'</code> and forwards payloads to Lambda.
AWS Lambda	Normalizes payloads, classifies worker condition, and coordinates storage or notification actions.
DynamoDB	Stores structured worker records using <code>DeviceID</code> and <code>Timestamp</code> .
Amazon S3	Stores historical JSON logs for traceability and later review.
Amazon SNS	Sends supervisor email notifications for <code>DANGER</code> and <code>EMERGENCY</code> events.

9.5 Connectivity Security and Cloud Configuration

The connectivity layer uses MQTT over TLS to transmit telemetry, HRV summaries, and alert events from the ESP32-C3 edge device to AWS IoT Core. This protects worker-related data during transmission and ensures that the device communicates with the cloud through an authenticated and encrypted channel.

The cloud configuration avoids hard-coding resource names directly inside the Lambda source code. Instead, the Lambda function uses environment variables, including `DDB_TABLE_NAME`, `S3_BUCKET_NAME`, and `SNS_TOPIC_ARN`, to define the DynamoDB table, S3 bucket, and SNS topic used by the system. In the submitted version, account-specific values such as `<ACCOUNT_ID>` are replaced with placeholders to prevent exposure of private AWS information.

The Lambda execution role follows a least-privilege configuration. It grants only the permissions required for the implemented cloud workflow: `dynamodb:PutItem` for storing processed records, `s3:PutObject` for saving historical JSON logs, and `sns:Publish` for sending supervisor notifications. This limits the cloud function to the minimum actions required for data storage and alert communication.

10 Demonstration Setup

This section describes the experimental setup and test scenarios used to validate the hybrid edge-cloud prototype. The demonstration verifies sensor data acquisition, local risk evaluation, buzzer activation, MQTT publishing, and cloud processing (AWS IoT Core, Lambda, DynamoDB, S3, SNS).

10.1 Hardware Testbench Configuration

The prototype uses a XIAO ESP32-C3 as the central edge device. The MAX30102 (I2C) provides heart rate and SpO_2 ; the MPU6050 (I2C) captures acceleration and gyroscope for fall detection. An active buzzer (GPIO2) gives local warnings, and a cancel button (GPIO3, pull-up) allows manual alarm silencing. The hardware is mounted on a mock wearable to simulate worker motion. $\text{PM}_{2.5}$ sensing and SIM768x-based 4G/SMS remain planned or simulated extensions (not physically validated in this testbench).

10.2 Experimental System Workflow

The live demonstration proceeds in four stages:

- (i) **Startup and connection:** The ESP32-C3 powers on, initialises I2C, starts the MAX30102 and MPU6050, connects to Wi-Fi, and establishes a TLS-secured MQTT connection to AWS IoT Core.
- (ii) **Edge monitoring:** FreeRTOS runs three tasks (anomaly detection, alarm control, cloud publishing). Shared data are mutex-protected; HRV summaries and alerts use queues.
- (iii) **Scenario testing:** Normal, danger, fall-emergency, and exhaustion-risk cases are simulated via controlled movement and JSON payloads. The edge checks thresholds and activates the buzzer when needed.
- (iv) **Cloud verification:** MQTT messages are inspected in the AWS IoT test client. The IoT Rule forwards them to the `WorkerDataProcessor`

Lambda, which classifies the worker state, stores records in DynamoDB, logs to S3, and sends SNS emails for danger or emergency.

10.3 Demonstration Scenarios and Test Vectors

Four scenarios were used to evaluate the expected edge–cloud behaviour. The test vectors represent the main risk states used in the prototype: normal monitoring, biometric danger, fall emergency, and exhaustion risk.

Table 10: Demonstration scenarios and observed system responses

Scenario	Input condition	Observed response
Normal state	HR = 82 BPM, SpO ₂ = 97%, SVM = 9.8, RMSSD = 35.2, fall = false	Edge alarm level = 0; cloud state = NORMAL ; data are stored without SNS email notification.
Biometric danger	HR = 118 BPM, SpO ₂ = 88%, SVM = 9.7, RMSSD = 16.2, fall = false	Edge warning/danger condition is generated; cloud state = DANGER ; SNS email notification is sent.
Fall emergency	HR = 126 BPM, SpO ₂ = 93%, SVM = 31.2, RMSSD = 14.1, fall = true	Edge emergency alarm is generated; cloud state = EMERGENCY ; SNS email notification is sent.
Exhaustion risk	HR = 116 BPM, SpO ₂ = 94%, SVM = 9.8, RMSSD = 13.8, fall = false	Fatigue-related emergency condition is classified because RMSSD < 15 ms and HR > 110 BPM; SNS email notification is sent.

10.4 Cloud Validation Outputs

The cloud-side validation checks whether each MQTT payload is correctly forwarded, classified, stored, and notified.

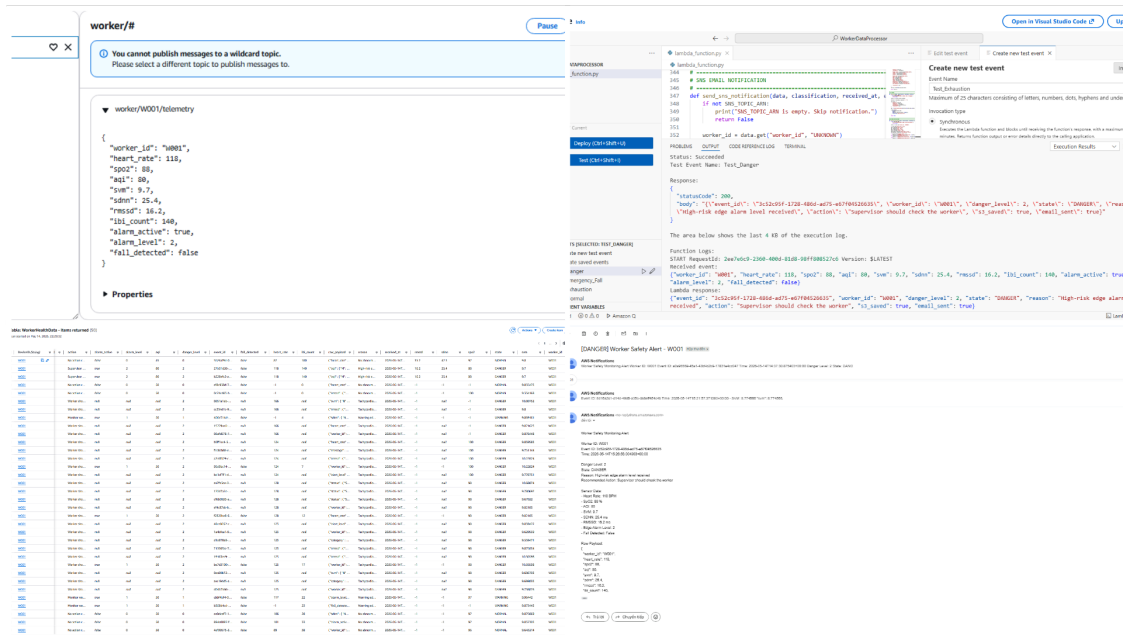


Figure 6: Cloud-side integration and validation outputs for the DANGER case verification

The expected validation evidence includes:

- successful MQTT message reception in the AWS IoT MQTT test client using the topic filter `worker/#`;
- Lambda response showing `danger_level`, `state`, `reason`, `action`, `s3_saved`, and `email_sent`;
- new processed records in the DynamoDB table `WorkerHealthData`;
- historical JSON logs stored in the configured S3 bucket;

- SNS email notification for DANGER and EMERGENCY cases only.

10.5 Experimental Results and Evaluation Matrix

Table 11 summarizes the demonstration test vectors and the observed edge–cloud results. The evaluation verifies that normal telemetry is logged without notification, while danger and emergency events are classified and escalated through the cloud notification pathway.

Table 11: Validation matrix of demonstration test files

Test file	HR	SpO ₂	SVM	RMSSD	Cloud state	SNS
Test_Normal.json	82	97	9.8	35.2	NORMAL	No
Test_Danger.json	118	88	9.7	16.2	DANGER	Yes
Test_Emergency_Fall.json	126	93	31.2	14.1	EMERGENCY	Yes
Test_Exhaustion.json	116	94	9.8	13.8	EMERGENCY	Yes

10.6 Demonstration Scope and Limitations

The demonstration validates the core edge–cloud workflow: sensor reading, local alarm, MQTT, Lambda classification, DynamoDB, S3, and SNS. However, it is a controlled testbench, not a field deployment. PM_{2.5} sensing is a placeholder; SIM768x-based notifications are simulated. HRV uses BPM-based IBI approximation; production should use beat-to-beat PPG timestamps and field tests with real workers.

11 Security, Safety and Ethics

Data security. TLS-secured MQTT protects worker data during transmission. Sensitive information (AWS identifiers, certificates, keys, credentials) is never hard-coded; placeholders are used in the submitted code, and secure configuration would be used in deployment.

Cloud access control. The Lambda execution role follows least privilege, granting only `dynamodb:PutItem`, `s3:PutObject`, and `sns:Publish`. This minimises exposure and supports safe cloud operation.

Worker privacy. Informed consent is required. Only authorised supervisors see identifiable alerts; data for analysis are anonymised. Collected data are limited to worker ID, heart rate, SpO₂, HRV, fall status, alarm level, and timestamp.

Physical safety. The prototype uses low-voltage electronics, a local buzzer, and a cancel button to silence false alarms. PM_{2.5} sensing and SIM768x-based 4G emergency communication remain planned or simulated extensions; their physical integration requires separate validation.

Ethical use. The system is designed solely for worker protection, not disciplinary surveillance. Policies must prohibit punitive use of HRV, fatigue, location, or health-risk data. Instead, data should support rest breaks, hydration, medical response, and safer work allocation.

12 Impact and Feasibility

Impact. The proposed system offers health, economic, and social benefits. Early warnings for heat stress and fatigue can help prevent heat stroke, falls, and related incidents on construction sites. Economically, fewer accidents reduce lost workdays, medical costs, and insurance burdens. Socially, the system demonstrates how accessible digital technology can improve protection for outdoor labourers in high-risk environments.

Feasibility. Technically, the system uses commercially available components (XIAO ESP32-C3, MAX30102, MPU6050) with mature software libraries. The ESP32-C3 supports sensor fusion, HRV computation, and rule-based classification. (WBGT heat-stress estimation is not part of the current prototype.) Operationally, routine telemetry requires only a standard Wi-Fi router; cellular connectivity can be added for emergency alerts. Financially, the prototype BOM of approximately US\$46 enables affordable pilot deployment, with further cost reductions expected at scale.

These factors indicate that the proposed solution is suitable for real-world testing and has strong potential for scalable deployment.

13 Limitations and Future Work

Several limitations remain. Battery life is approximately 8–10 hours, insufficient for extended shifts without a larger battery or aggressive duty cycling. WBGT estimation without a black-globe thermometer may underestimate radiative heat load; a globe sensor should be integrated for compliance-grade assessment. HRV thresholds are literature-based; personalised baselines would improve fatigue detection. Finally, the enclosure needs optimisation for comfort, durability, and impact resistance.

Future work includes extending autonomy through flexible solar charging and thin-film batteries, personalising fatigue detection by learning individual HRV baselines, and adding real-time location tracking (GNSS/UWB) to map incidents onto BIM platforms. Blockchain-based safety logs may also be explored for transparent labour inspection and SDG reporting. Longitudinal field studies across multiple construction sites are needed to quantify effects on injury rates, worker health, and productivity.

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